Title: **QVI Data Analytics Virtual Experience**

Task 1: Customer Analytics

R code and Analysis

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Executive Summary

This analysis examined customer purchasing behavior using transaction and customer datasets provided by QVI. The objective was to identify purchasing trends, evaluate the key drivers of sales performance, and provide data-driven recommendations to support the client's marketing and merchandising strategy.

The datasets were cleaned by removing duplicate records, checking for missing values, excluding non-chip products (such as salsa), and engineering additional variables including brand and pack size. The cleaned datasets were then merged to enable customer segmentation and sales analysis.

The analysis revealed that Older Singles/Couples generated the highest total sales, while Retirees represented the largest customer base. Mainstream customers contributed the highest overall revenue among customer types. Product analysis showed that Kettle was the highest-performing brand across customer segments, and 175g was the most popular pack size in terms of sales and transaction volume. Family customer segments also purchased a higher average number of units per transaction, suggesting opportunities for bundle promotions.

Based on these findings, it is recommended that the client prioritize inventory for Kettle products and 175g pack sizes, develop targeted marketing campaigns for Older Singles/Couples and Retirees, and introduce promotional offers aimed at family households to encourage larger basket sizes. These strategies are expected to strengthen customer engagement, improve inventory planning, and increase overall sales performance.

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1. Introduction

The purpose of this project is to analyze customer transaction and purchasing behavior using data provided by QVI. Understanding customer preferences, purchasing patterns, and sales performance enables businesses to make informed decisions regarding product assortment, inventory management, and targeted marketing strategies. This report describes the data preparation process, exploratory data analysis, feature engineering, visualization of key findings, and business recommendations based on the results.

2. Load Libraries

```r
#Installing packages
> install.packages(c(
+     "tidyverse",
+     "readxl",
+     "psych",
+     "skimr",
+     "janitor",
+     "car",
+     "GGally",
+     "corrplot",
+     "ggpubr",
+     "rstatix",
+     "effectsize"
+ ))
#Loading required libraries
library(tidyverse)
library(readxl)
library(stringr)
library(scales)
```

3. Import Data

```
#Importing datasets


data1 <- read.csv(file.choose())

> names(data1)

[1] "LYLTY_CARD_NBR"   "LIFESTAGE"        "PREMIUM_CUSTOMER"

> head(data1)

  LYLTY_CARD_NBR              LIFESTAGE PREMIUM_CUSTOMER

1           1000  YOUNG SINGLES/COUPLES          Premium

2           1002  YOUNG SINGLES/COUPLES       Mainstream

3           1003         YOUNG FAMILIES           Budget

4           1004  OLDER SINGLES/COUPLES       Mainstream

5           1005 MIDAGE SINGLES/COUPLES       Mainstream

6           1007  YOUNG SINGLES/COUPLES           Budget

> data2<- read_excel(file.choose())

> head(data2)

# A tibble: 6 × 8

   DATE STORE_NBR LYLTY_CARD_NBR TXN_ID PROD_NBR PROD_NAME
PROD_QTY TOT_SALES

  <dbl>     <dbl>          <dbl>  <dbl>    <dbl> <chr>
<dbl>     <dbl>

1 43390         1           1000      1        5 Natural Chip        Compny SeaSalt175g
2       6

2 43599         1           1307    348       66 CCs Nacho Cheese    175g
3       6.3

3 43605         1           1343    383       61 Smiths Crinkle Cut  Chips Chicken 170g
2       2.9

4 43329         2           2373    974       69 Smiths Chip Thinly  S/Cream&Onion 175g
5      15

5 43330         2           2426   1038      108 Kettle Tortilla ChpsHny&Jlpno Chili 150g
3      13.8

6 43604         4           4074   2982       57 Old El Paso Salsa   Dip Tomato Mild 300g
1       5.1
```

4. Data Cleaning

```
# Checking missing values

> colSums(is.na(data1))

  LYLTY_CARD_NBR        LIFESTAGE PREMIUM_CUSTOMER

               0                0                0

> colSums(is.na(data2))

          DATE      STORE_NBR LYLTY_CARD_NBR         TXN_ID       PROD_NBR      PROD_NAME
PROD_QTY      TOT_SALES

             0              0              0              0              0              0
0              0


#Checking duplicate values

> sum(duplicated(data1))

[1] 0

> sum(duplicated(data1$LYLTY_CARD_NBR))

[1] 0

> sum(duplicated(data2))

[1] 1

> data2[duplicated(data2),]

# A tibble: 1 × 8

   DATE STORE_NBR LYLTY_CARD_NBR TXN_ID PROD_NBR PROD_NAME
PROD_QTY TOT_SALES

  <dbl>     <dbl>          <dbl>  <dbl>    <dbl> <chr>
<dbl>     <dbl>

1 43374       107         107024 108462       45 Smiths Thinly Cut   Roast Chicken 175g
2         6

#Deleting the duplicate

> data2 <- data2[!duplicated(data2),]

> sum(duplicated(data2))

[1] 0

#Customer segments

> table(data1$LIFESTAGE)


MIDAGE SINGLES/COUPLES           NEW FAMILIES         OLDER FAMILIES  OLDER SINGLES/COUPLES
RETIREES

                  7275                   2549                   9780                  14609
14805

        YOUNG FAMILIES  YOUNG SINGLES/COUPLES
```

```
                  9178                  14441
> table(data1$PREMIUM_CUSTOMER)


    Budget Mainstream    Premium
     24470      29245      18922
```

> #blank values

```
> colSums(data1 == "", na.rm = TRUE)
  LYLTY_CARD_NBR        LIFESTAGE  PREMIUM_CUSTOMER
               0                0                 0
> colSums(data2 == "", na.rm = TRUE)
        DATE      STORE_NBR  LYLTY_CARD_NBR         TXN_ID        PROD_NBR       PROD_NAME
PROD_QTY      TOT_SALES
             0              0              0              0              0              0
0              0
> sum(data1$LIFESTAGE == "", na.rm = TRUE)
[1] 0
```

> #convert the date

```
> data2$DATE <- as.Date(data2$DATE, origin = "1899-12-30")
> head(data2$DATE)
[1] "2018-10-17" "2019-05-14" "2019-05-20" "2018-08-17" "2018-08-18" "2019-05-19"
```

> #Examine product quantities

```
> summary(data2$PROD_QTY)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.000   2.000   2.000   1.907   2.000 200.000
```

> #outlier transaction(s)

> #maximum product quantity is an outlier because it suggests a commercial transaction. The task is to analyze retail transactions thus this is an outlier.

> #Removing the outlier

```
> data2[data2$PROD_QTY == 200, ]
# A tibble: 2 × 8
  DATE       STORE_NBR LYLTY_CARD_NBR TXN_ID PROD_NBR PROD_NAME
PROD_QTY TOT_SALES
  <date>         <dbl>          <dbl>  <dbl>    <dbl> <chr>
<dbl>     <dbl>
1 2018-08-19       226         226000 226201        4 Dorito Corn Chp     Supreme 380g
200       650
```

```
2 2019-05-20       226         226000 226210        4 Dorito Corn Chp     Supreme 380g
200       650

> #Now that the customer is not a consumer. the best option is to remove the customer
with that particular id

> data2 <- subset(data2, LYLTY_CARD_NBR != 226000)

> summary(data2$PROD_QTY)

   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

  1.000   2.000   2.000   1.906   2.000   5.000

> #Examine Sales

> summary(data2$TOT_SALES)

   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

  1.500   5.400   7.400   7.299   9.200  29.500

> #Examine product names

> summary(data2$PROD_NAME)

   Length     Class      Mode

   264833 character character

> #Check the salsa products

> #Because we are only tasked to analyze chips products, we can remove non-chips
products

> data2 %>% filter(str_detect(PROD_NAME,regex("salsa", ignore_case = TRUE)))

# A tibble: 18,094 × 8

   DATE       STORE_NBR LYLTY_CARD_NBR TXN_ID PROD_NBR PROD_NAME
PROD_QTY TOT_SALES

   <date>         <dbl>          <dbl>  <dbl>    <dbl> <chr>
<dbl>     <dbl>

 1 2019-05-19         4           4074   2982       57 Old El Paso Salsa   Dip Tomato Mild
300g          1       5.1

 2 2019-05-15        39          39144  35506       57 Old El Paso Salsa   Dip Tomato Mild
300g          1       5.1

 3 2019-05-20        45          45127  41122       64 Red Rock Deli SR    Salsa & Mzzrlla
150g          2       5.4

 4 2018-08-18        56          56013  50090       39 Smiths Crinkle Cut  Tomato Salsa 150g
1       2.6

 5 2019-05-15        82          82480  82047      101 Doritos Salsa       Medium 300g
1       2.6

 6 2018-08-15        94          94233  93956       65 Old El Paso Salsa   Dip Chnky Tom
Ht300g        1       5.1

 7 2018-08-16        97          97159  97271       35 Woolworths Mild     Salsa 300g
5       7.5
```

```
 8 2018-08-15       116         116184 120270       59 Old El Paso Salsa   Dip Tomato Med 300g
1       5.1

 9 2018-08-16       157         157185 159562       59 Old El Paso Salsa   Dip Tomato Med 300g
2      10.2

10 2018-08-19       175         175306 176634       59 Old El Paso Salsa   Dip Tomato Med 300g
1       5.1
```

```
#Removal of the non-chips products

> data2 <- data2 %>% filter(!str_detect(PROD_NAME, regex("salsa", ignore_case = TRUE)))

> data2 %>% filter(str_detect(PROD_NAME,regex("salsa", ignore_case = TRUE)))
```

5. Data Merging and Feature Engineering

```
# Data merging
> data3 <- left_join(data2, data1, by = "LYLTY_CARD_NBR")
> dim(data3)
[1] 246739     10
> str(data3)
tibble [246,739 × 10] (S3: tbl_df/tbl/data.frame)
$ DATE            : Date[1:246739], format: "2018-10-17" "2019-05-14" "2019-05-20" "2018-08-17" ...
$ STORE_NBR       : num [1:246739] 1 1 1 2 2 4 4 5 7 7 ...
$ LYLTY_CARD_NBR  : num [1:246739] 1000 1307 1343 2373 2426 ...
$ TXN_ID          : num [1:246739] 1 348 383 974 1038 ...
$ PROD_NBR        : num [1:246739] 5 66 61 69 108 16 24 42 52 16 ...
$ PROD_NAME       : chr [1:246739] "Natural Chip        Compny SeaSalt175g" "CCs Nacho Cheese 175g" "Smiths Crinkle Cut  Chips Chicken 170g" "Smiths Chip Thinly  S/Cream&Onion 175g" ...
$ PROD_QTY        : num [1:246739] 2 3 2 5 3 1 1 1 2 1 ...
$ TOT_SALES       : num [1:246739] 6 6.3 2.9 15 13.8 5.7 3.6 3.9 7.2 5.7 ...
$ LIFESTAGE       : chr [1:246739] "YOUNG SINGLES/COUPLES" "MIDAGE SINGLES/COUPLES" "MIDAGE SINGLES/COUPLES" "MIDAGE SINGLES/COUPLES" ...
$ PREMIUM_CUSTOMER: chr [1:246739] "Premium" "Budget" "Budget" "Budget" ...
> colSums(is.na(data3))
DATE        STORE_NBR    LYLTY_CARD_NBR           TXN_ID          PROD_NBR         PROD_NAME
PROD_QTY
0                0                0                0                0                0
0
TOT_SALES        LIFESTAGE PREMIUM_CUSTOMER
0                0                0


#Saving the 2 cleaned datasets and the merged data into a single file
save(data1,data2, data3, file = "QVI_Cleaned_merged_data.RData")


> #Feature extraction
> data3$PACK_SIZE <- as.numeric(str_extract(data3$PROD_NAME, "\\d+"))
> summary(data3$PACK_SIZE)
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
70.0   150.0   170.0   175.6   175.0   380.0
```

```
> data3$BRAND <- word(data3$PROD_NAME, 1)

> table(data3$BRAND)


Burger        CCs    Cheetos   Cheezels       Cobs     Dorito    Doritos     French      Grain
GrnWves  Infuzions

1564       4551       2927       4603       9693       3183      22041       1418       6272
1468      11057

Infzns     Kettle    Natural        NCC   Pringles        Red        RRD      Smith     Smiths
Snbts   Sunbites

3144      41288       6050       1419      25102       4427      11894       2963      27389
1576       1432

Thins   Tostitos   Twisties   Tyrrells Woolworths         WW

14075       9471       9454       6442       1516      10320
```

6. Exploratory Data Analysis

```
> #Calculate Key Performance Metrics
> #Total sales
> sum(data3$TOT_SALES)
[1] 1805172
> #Total Transactions
> nrow(data3)
[1] 246739
> #Total customers
> n_distinct(data3$LYLTY_CARD_NBR)
[1] 71287
> #Average Transaction Value
> mean(data3$TOT_SALES)
[1] 7.316118
> #Average Quantity Purchased
> mean(data3$PROD_QTY)
[1] 1.906456
> #Average Units per Customer
> data3 %>% group_by(LYLTY_CARD_NBR) %>% summarise(units = sum(PROD_QTY)) %>% summarise(Avg = mean(units))
# A tibble: 1 × 1
    Avg
  <dbl>
1  6.60
> #Sales by customer Segment
> sales_segment <- data3 %>%
+     group_by(LIFESTAGE, PREMIUM_CUSTOMER) %>%
+     summarise(
+         Total_Sales = sum(TOT_SALES),
+         Transactions = n(),
+         Customers = n_distinct(LYLTY_CARD_NBR),
+         Avg_Spend = mean(TOT_SALES),
+         .groups = "drop"
+     )
> sales_segment
```

```
# A tibble: 21 × 6
   LIFESTAGE              PREMIUM_CUSTOMER Total_Sales Transactions Customers Avg_Spend
   <chr>                  <chr>                  <dbl>        <int>     <int>     <dbl>
 1 MIDAGE SINGLES/COUPLES Budget                33346.         4691      1474      7.11
 2 MIDAGE SINGLES/COUPLES Mainstream            84734.        11095      3298      7.64
 3 MIDAGE SINGLES/COUPLES Premium               54444.         7612      2369      7.15
 4 NEW FAMILIES           Budget                20607.         2824      1087      7.30
 5 NEW FAMILIES           Mainstream            15980.         2185       830      7.31
 6 NEW FAMILIES           Premium               10761.         1488       575      7.23
 7 OLDER FAMILIES         Budget               156864.        21514      4611      7.29
 8 OLDER FAMILIES         Mainstream            96414.        13241      2788      7.28
 9 OLDER FAMILIES         Premium               75243.        10403      2231      7.23
10 OLDER SINGLES/COUPLES  Budget               127834.        17172      4849      7.44
> #Sales Drivers
> #Which life stage spends the most
> data3 %>% group_by(LIFESTAGE) %>% summarise(Sales = sum(TOT_SALES)) %>% arrange(desc(Sales))
# A tibble: 7 × 2
  LIFESTAGE                Sales
  <chr>                    <dbl>
1 OLDER SINGLES/COUPLES  376014.
2 RETIREES               342382.
3 OLDER FAMILIES         328520.
4 YOUNG FAMILIES         294628.
5 YOUNG SINGLES/COUPLES  243757.
6 MIDAGE SINGLES/COUPLES 172524.
7 NEW FAMILIES            47348.
> #Which customer type spends most
> data3 %>% group_by(PREMIUM_CUSTOMER) %>% summarise(Sales = sum(TOT_SALES)) %>%
arrange(desc(Sales))
# A tibble: 3 × 2
  PREMIUM_CUSTOMER   Sales
  <chr>              <dbl>
1 Mainstream       700865.
2 Budget           631407.
3 Premium          472899.
```

```
> #Top selling brands
> data3 %>% group_by(BRAND) %>% summarise(Sales = sum(TOT_SALES)) %>% arrange(desc(Sales))
# A tibble: 28 × 2
   BRAND       Sales
   <chr>       <dbl>
 1 Kettle    390240.
 2 Smiths    202903.
 3 Doritos   187278.
 4 Pringles  177656.
 5 Thins      88853.
 6 Twisties   81522.
 7 Tostitos   79790.
 8 Infuzions  76248.
 9 Cobs       70570.
10 RRD        64954.
#Most popular pack sizes
> data3 %>% group_by(PACK_SIZE) %>% summarise(Sales = sum(TOT_SALES)) %>% arrange(desc(Sales))
# A tibble: 20 × 2
   PACK_SIZE   Sales
       <dbl>   <dbl>
 1       175 485431.
 2       150 289682.
 3       134 177656.
 4       110 162765.
 5       170 146673
 6       330 136794.
 7       165 101361.
 8       380  75420.
 9       270  55425.
10       210  43049.
11       250  26097.
12       135  26090.
13       200  16008.
14       190  14413.
15       160  10648.
```

```
16        90   9676.
17       180   8568.
18        70   6852
19       220   6831
20       125   5733
> data3 %>%
+     group_by(PREMIUM_CUSTOMER) %>%
+     summarise(
+         Avg_Spend = mean(TOT_SALES),
+         Transactions = n(),
+         Customers = n_distinct(LYLTY_CARD_NBR)
+     )
# A tibble: 3 × 4
  PREMIUM_CUSTOMER Avg_Spend Transactions Customers
  <chr>                <dbl>        <int>     <int>
1 Budget                7.28        86762     24006
2 Mainstream            7.37        95043     28734
3 Premium               7.28        64934     18547
```

> #In the customer type spending behavior; Mainstream customers are the strongest-performing customer segment, generating the highest total sales. This performance appears to be driven by a combination of having the largest customer base, the highest transaction volume, and a slightly higher average spend per transction compared with budget and premium customers.

```
>
> #Lifestage
> data3 %>% group_by(LIFESTAGE) %>% summarise(Avg_Spend = mean(TOT_SALES), Transactions = n(), Customers = n_distinct(LYLTY_CARD_NBR))
# A tibble: 7 × 4
  LIFESTAGE              Avg_Spend Transactions Customers
  <chr>                      <dbl>        <int>     <int>
1 MIDAGE SINGLES/COUPLES      7.37        23398      7141
2 NEW FAMILIES                7.29         6497      2492
3 OLDER FAMILIES              7.27        45158      9630
4 OLDER SINGLES/COUPLES       7.40        50792     14389
5 RETIREES                    7.37        46431     14555
6 YOUNG FAMILIES                7.28        40494      9036
7 YOUNG SINGLES/COUPLES       7.18        33969     14044
```

```
> data3 %>% group_by(LIFESTAGE) %>% summarise(Avg_Spend = mean(TOT_SALES), Transactions = n(), 
Customers = n_distinct(LYLTY_CARD_NBR)) %>% arrange(desc(Customers))

# A tibble: 7 × 4

  LIFESTAGE              Avg_Spend Transactions Customers

  <chr>                      <dbl>        <int>     <int>

1 RETIREES                    7.37        46431     14555

2 OLDER SINGLES/COUPLES       7.40        50792     14389

3 YOUNG SINGLES/COUPLES       7.18        33969     14044

4 OLDER FAMILIES              7.27        45158      9630

5 YOUNG FAMILIES              7.28        40494      9036

6 MIDAGE SINGLES/COUPLES      7.37        23398      7141

7 NEW FAMILIES                7.29         6497      2492

> data3 %>% group_by(LIFESTAGE) %>% summarise(Avg_Spend = mean(TOT_SALES), Transactions = n(), 
Customers = n_distinct(LYLTY_CARD_NBR), Sales = sum(TOT_SALES)) %>% arrange(desc(Sales))

# A tibble: 7 × 5

  LIFESTAGE              Avg_Spend Transactions Customers   Sales

  <chr>                      <dbl>        <int>     <int>   <dbl>

1 OLDER SINGLES/COUPLES       7.40        50792     14389 376014.

2 RETIREES                    7.37        46431     14555 342382.

3 OLDER FAMILIES              7.27        45158      9630 328520.

4 YOUNG FAMILIES              7.28        40494      9036 294628.

5 YOUNG SINGLES/COUPLES       7.18        33969     14044 243757.

6 MIDAGE SINGLES/COUPLES      7.37        23398      7141 172524.

7 NEW FAMILIES                7.29         6497      2492  47348.

> data3 %>%

+     group_by(LIFESTAGE) %>%

+     summarise(

+         Avg_Units = mean(PROD_QTY),

+         Avg_Spend = mean(TOT_SALES),

+         Transactions = n(),

+         Customers = n_distinct(LYLTY_CARD_NBR),

+         Sales = sum(TOT_SALES)

+     ) %>%

+     arrange(desc(Avg_Units))

# A tibble: 7 × 6

  LIFESTAGE              Avg_Units Avg_Spend Transactions Customers   Sales
```

```
  <chr>                        <dbl>     <dbl>        <int>     <int>    <dbl>
1 OLDER FAMILIES                1.95      7.27        45158      9630 328520.
2 YOUNG FAMILIES                1.94      7.28        40494      9036 294628.
3 OLDER SINGLES/COUPLES         1.91      7.40        50792     14389 376014.
4 MIDAGE SINGLES/COUPLES        1.90      7.37        23398      7141 172524.
5 RETIREES                      1.89      7.37        46431     14555 342382.
6 NEW FAMILIES                  1.86      7.29         6497      2492  47348.
7 YOUNG SINGLES/COUPLES         1.83      7.18        33969     14044 243757.
```

> #From the analysis: Older singles/Couples are the highest-value segment beacause they generate the most revenue, make the most transactions, and spend the most per transaction. Older families and Young Families tend to purchase the highest number of chip packs per visit, suggesting family households buy in large quantities.Retirees contribute substantially to revenue due to their large customes and frequent purchase. Differences in total sales are influenced by both customer numbers and purchase frequency, while average spend per transaction varies only slightly across life stages.

```
>
> # brand Vs sales behavior
> brand_sales <- data3 %>%
+   group_by(BRAND) %>%
+   summarise(
+     Total_Sales = sum(TOT_SALES),
+     Transactions = n(),
+     Avg_Units = mean(PROD_QTY),
+     .groups = "drop"
+   ) %>%
+   arrange(desc(Total_Sales))
> brand_sales
# A tibble: 28 × 4
   BRAND      Total_Sales Transactions Avg_Units
   <chr>            <dbl>        <int>     <dbl>
 1 Kettle         390240.        41288      1.91
 2 Smiths         202903.        27389      1.90
 3 Doritos        187278.        22041      1.92
 4 Pringles       177656.        25102      1.91
 5 Thins           88853.        14075      1.91
 6 Twisties        81522.         9454      1.92
 7 Tostitos        79790.         9471      1.91
```

```
 8 Infuzions      76248.        11057      1.91
 9 Cobs           70570.         9693      1.92
10 RRD            64954.        11894      1.89
```

> #Kettle is the clear market leader, generating almost twice the sales of the second-ranked brand(Smiths).Smiths, Doritos and Pringles also contribute substantially to total revenue. The average units per transaction are very similar across brands (around 1.9 units), indicating that customers tend to purchase about packs regardless of brand. This susggest that differences in total sales are driven primarily by how often ccustomes buy each brand rather than by purchasing larger quantities of one brand over the other.

```
>
> #Most popular pack_sizes
> data3 %>% group_by(PACK_SIZE) %>% summarise(Sales = sum(TOT_SALES), Transactions = n(), Avg_Units = mean(PROD_QTY), Customers = n_distinct(LYLTY_CARD_NBR), Avg_spend = mean(TOT_SALES)) %>% arrange(desc(Sales))
# A tibble: 20 × 6
   PACK_SIZE   Sales Transactions Avg_Units Customers Avg_spend
       <dbl>   <dbl>        <int>     <dbl>     <int>     <dbl>
 1       175 485431.        66389      1.90     40736      7.31
 2       150 289682.        40203      1.91     29359      7.21
 3       134 177656.        25102      1.91     20657      7.08
 4       110 162765.        22387      1.91     18678      7.27
 5       170 146673         19983      1.91     16916      7.34
 6       330 136794.        12540      1.91     11282     10.9
 7       165 101361.        15297      1.90     13229      6.63
 8       380  75420.         6416      1.91      6095     11.8
 9       270  55425.         6285      1.92      5924      8.82
10       210  43049.         6272      1.91      5921      6.86
11       250  26097.         3169      1.92      3092      8.23
12       135  26090.         3257      1.91      3179      8.01
13       200  16008.         4473      1.88      4165      3.58
14       190  14413.         2995      1.89      2874      4.81
15       160  10648.         2970      1.89      2841      3.59
16        90   9676.         3008      1.89      2875      3.22
17       180   8568.         1468      1.88      1440      5.84
18        70   6852          1507      1.89      1474      4.55
19       220   6831          1564      1.90      1519      4.37
20       125   5733          1454      1.88      1416      3.94
```

```
> #175g is by far the most popular pack size with the highest total sales, highest number of transactions and the largest number of customers purchasing it. 150g is the second most popular pack size. 330g and 380g stand out because they have much higher average spend per transaction (10.9 and 11.8 respectively), which is expected since they are larger pack. However, they generate fewer transactions than the smaller packs. The average units purchased per transaction remain close to 1.9 across all pack sizes, suggesting customers typically buy about 2 packs regardless of size.

>

> #Busines implications: Ensure 175g and 150g packs remain well stocked, as they drive the largest share of sales.Consider promotions or targeted marketing around these popular pack sizes because they appeal to the widest customer base. Larger packs (330g and 380g) generate higher spend per transaction and may be suitable for promotions aimed at family shoppers or larger households.

>

> #Lifestage and customer type buy the most 175g packs
> data3 %>%
+     filter(PACK_SIZE == 175) %>%
+     group_by(LIFESTAGE, PREMIUM_CUSTOMER) %>%
+     summarise(
+         Sales = sum(TOT_SALES),
+         Transactions = n(),
+         .groups = "drop"
+     ) %>%
+     arrange(desc(Sales))
# A tibble: 21 × 4
   LIFESTAGE              PREMIUM_CUSTOMER  Sales Transactions
   <chr>                  <chr>             <dbl>        <int>
 1 OLDER FAMILIES         Budget           42205.         5808
 2 RETIREES               Mainstream       38243.         5295
 3 YOUNG SINGLES/COUPLES Mainstream       37968.         4997
 4 YOUNG FAMILIES         Budget           35635.         4921
 5 OLDER SINGLES/COUPLES Budget           34497          4625
 6 OLDER SINGLES/COUPLES Premium          33387.         4457
 7 OLDER SINGLES/COUPLES Mainstream       33042.         4525
 8 RETIREES               Budget           28977.         3847
 9 OLDER FAMILIES         Mainstream       25975.         3588
10 RETIREES               Premium          24868.         3306
#The result insights: Budget Older Families are the strongest buyers of the most popular pack size (175g, making them an attractive segment for promotions on standard-sized packs. Mainstream Retirees and Mainstream YOung Singles/Couples are also major purchasers of 175g
```

packs. Several Older Single/Couples segments (Budget, Premium, and Mainstream) appear in the top rankings, reinforcing the earlier finding that this stage is highly valuable. Finally, this suggests that the 175g pack appeals across multiple customer groups, but particularly to Older Families and Older Singles/Couples

```
>
> # Favorite brand for each segment
> brand_preference <- data3 %>%
+     group_by(LIFESTAGE, PREMIUM_CUSTOMER, BRAND) %>%
+     summarise(
+         Sales = sum(TOT_SALES),
+         Transactions = n(),
+         .groups = "drop"
+     ) %>%
+     arrange(LIFESTAGE, PREMIUM_CUSTOMER, desc(Sales))
> brand_preference
# A tibble: 588 × 5
   LIFESTAGE              PREMIUM_CUSTOMER BRAND     Sales Transactions
   <chr>                  <chr>            <chr>     <dbl>        <int>
 1 MIDAGE SINGLES/COUPLES Budget           Kettle    6736.          713
 2 MIDAGE SINGLES/COUPLES Budget           Smiths    3708.          522
 3 MIDAGE SINGLES/COUPLES Budget           Doritos   3556.          419
 4 MIDAGE SINGLES/COUPLES Budget           Pringles  3160.          449
 5 MIDAGE SINGLES/COUPLES Budget           Thins     1673.          269
 6 MIDAGE SINGLES/COUPLES Budget           Twisties  1505.          176
 7 MIDAGE SINGLES/COUPLES Budget           Tostitos  1404.          163
 8 MIDAGE SINGLES/COUPLES Budget           Infuzions 1399.          202
 9 MIDAGE SINGLES/COUPLES Budget           RRD       1342.          252
10 MIDAGE SINGLES/COUPLES Budget           Cobs      1311           180
top_brand <- brand_preference %>%
+     group_by(LIFESTAGE, PREMIUM_CUSTOMER) %>%
+     slice_max(Sales, n = 1, with_ties = FALSE) %>%
+     arrange(desc(Sales))
> top_brand
# A tibble: 21 × 5
# Groups:   LIFESTAGE, PREMIUM_CUSTOMER [21]
   LIFESTAGE              PREMIUM_CUSTOMER BRAND   Sales Transactions
```

```
  <chr>                  <chr>            <chr>   <dbl>         <int>
 1 YOUNG SINGLES/COUPLES  Mainstream       Kettle 35424.         3844
 2 OLDER FAMILIES         Budget           Kettle 32058          3320
 3 RETIREES               Mainstream       Kettle 31652.         3386
 4 OLDER SINGLES/COUPLES  Budget           Kettle 29066.         3065
 5 OLDER SINGLES/COUPLES  Premium          Kettle 27943.         2947
 6 OLDER SINGLES/COUPLES  Mainstream       Kettle 26853.         2835
 7 YOUNG FAMILIES         Budget           Kettle 26370.         2743
 8 RETIREES               Budget           Kettle 24340          2592
 9 RETIREES               Premium          Kettle 20922.         2216
10 MIDAGE SINGLES/COUPLES Mainstream       Kettle 20232.         2136
```

#Kettle is top-selling brand in every customer segment shown in your top 10 results. It consistently generates the highest sales across different life stages and customer types. This strongly suggests that Kettle has broad appeal, regardless of wheather customers are Budget, Mainstream, or Premium.

7. Visualizations

```
> # Summarize total sales by life stage
  > life_stage_sales <- data3 %>%
  +     group_by(LIFESTAGE) %>%
  +     summarise(
    +         Total_Sales = sum(TOT_SALES),
    +         .groups = "drop"
    +     )
>
  > # Create the chart
  > ggplot(life_stage_sales,
           +        aes(x = reorder(LIFESTAGE, Total_Sales),
                        +            y = Total_Sales)) +
  +     geom_col(fill = "steelblue") +
  +     geom_text(aes(label = comma(round(Total_Sales))),
                  +               hjust = -0.1,
                  +               size = 4) +
  +     coord_flip() +
  +     scale_y_continuous(
    +         labels = comma,
    +         expand = expansion(mult = c(0, 0.08))
    +     ) +
  +     labs(
    +         title = "Total Sales by Customer Life Stage",
    +         x = "Life Stage",
    +         y = "Total Sales ($)"
    +     ) +
  +     theme_minimal() +
  +     theme(
    +         plot.title = element_text(
      +             hjust = 0.5,
      +             face = "bold",
      +             size = 16
      +         ),
    +         axis.title = element_text(
```

```
+             face = "bold",
+             size = 12
+         ),
+         axis.text = element_text(size = 11)
+     )
>
> #Total Sales by Customer Type
> customer_sales <- data3 %>%
+     group_by(PREMIUM_CUSTOMER) %>%
+     summarise(Total_Sales = sum(TOT_SALES), .groups = "drop")
>
> ggplot(customer_sales,
+        aes(x = reorder(PREMIUM_CUSTOMER, Total_Sales),
+            y = Total_Sales)) +
+     geom_col(fill = "steelblue") +
+     geom_text(aes(label = comma(round(Total_Sales))),
+               vjust = -0.4,
+               size = 4) +
+     scale_y_continuous(labels = comma,
+                        expand = expansion(mult = c(0,0.08))) +
+     labs(
+         title = "Total Sales by Customer Type",
+         x = "Customer Type",
+         y = "Total Sales ($)"
+     ) +
+     theme_minimal() +
+     theme(
+         plot.title = element_text(face="bold",hjust=.5,size=16),
+         axis.title = element_text(face="bold")
+     )
>
> #Top 10 brands by Total sales
> top10_brands <- brand_sales %>%
+     slice_max(Total_Sales, n = 10)
>
```

```
> ggplot(top10_brands,
+        aes(x = reorder(BRAND, Total_Sales),
+            y = Total_Sales)) +
+     geom_col(fill = "steelblue") +
+     geom_text(aes(label = comma(round(Total_Sales))),
+               hjust = -0.1,
+               size = 3.8) +
+     coord_flip() +
+     scale_y_continuous(labels = comma,
+                        expand = expansion(mult = c(0,0.08))) +
+     labs(
+         title = "Top 10 Brands by Total Sales",
+         x = "Brand",
+         y = "Total Sales ($)"
+     ) +
+     theme_minimal() +
+     theme(
+         plot.title = element_text(face="bold",hjust=.5,size=16),
+         axis.title = element_text(face="bold")
+     )
>
> #Top 10 Pack Sizes
> pack_sales <- data3 %>%
+     group_by(PACK_SIZE) %>%
+     summarise(
+         Total_Sales = sum(TOT_SALES),
+         .groups = "drop"
+     ) %>%
+     arrange(desc(Total_Sales))
>
> top10_pack <- pack_sales %>%
+     slice_max(Total_Sales, n = 10)
>
> ggplot(top10_pack,
+        aes(x = reorder(as.factor(PACK_SIZE), Total_Sales),
```

```
+             y = Total_Sales)) +
+     geom_col(fill = "steelblue") +
+     geom_text(aes(label = comma(round(Total_Sales))),
+               vjust = -0.4,
+               size = 3.8) +
+     scale_y_continuous(labels = comma,
+                        expand = expansion(mult = c(0,0.08))) +
+     labs(
+         title = "Top 10 Pack Sizes by Total Sales",
+         x = "Pack Size (g)",
+         y = "Total Sales ($)"
+     ) +
+     theme_minimal() +
+     theme(
+         plot.title = element_text(face="bold",hjust=.5,size=16),
+         axis.title = element_text(face="bold")
+     )
>
> #Top brand by Customer Segment
> ggplot(top_brand,
+        aes(x = reorder(paste(LIFESTAGE,
+                              PREMIUM_CUSTOMER,
+                              sep = " - "),
+                        Sales),
+            y = Sales)) +
+     geom_col(fill = "steelblue") +
+     geom_text(aes(label = comma(round(Sales))),
+               hjust = -0.1,
+               size = 3.5) +
+     coord_flip() +
+     scale_y_continuous(labels = comma,
+                        expand = expansion(mult = c(0,0.08))) +
+     labs(
+         title = "Top Brand by Customer Segment",
+         x = "Customer Segment",
```

```
+         y = "Total Sales ($)"
+     ) +
+     theme_minimal() +
+     theme(
+         plot.title = element_text(face="bold",hjust=.5,size=16),
+         axis.title = element_text(face="bold"),
+         axis.text.y = element_text(size=8)
+     )
>
> #Average Spend by Customer Type
> avg_spend <- data3 %>%
+     group_by(PREMIUM_CUSTOMER) %>%
+     summarise(
+         Avg_Spend = mean(TOT_SALES),
+         .groups = "drop"
+     )
>
> ggplot(avg_spend,
+        aes(x = PREMIUM_CUSTOMER,
+             y = Avg_Spend)) +
+     geom_col(fill = "steelblue") +
+     geom_text(aes(label = round(Avg_Spend,2)),
+               vjust = -0.4,
+               size = 4) +
+     labs(
+         title = "Average Spend per Transaction by Customer Type",
+         x = "Customer Type",
+         y = "Average Spend ($)"
+     ) +
+     theme_minimal() +
+     theme(
+         plot.title = element_text(face="bold",hjust=.5,size=16),
+         axis.title = element_text(face="bold")
+     )
>
```

```
> #Average Units purchased by Life Stage
> avg_units <- data3 %>%
+     group_by(LIFESTAGE) %>%
+     summarise(
  +         Avg_Units = mean(PROD_QTY),
  +         .groups = "drop"
  +     )
>
> ggplot(avg_units,
         +        aes(x = reorder(LIFESTAGE, Avg_Units),
                      +            y = Avg_Units)) +
+     geom_col(fill = "steelblue") +
+     geom_text(aes(label = round(Avg_Units,2)),
                +               hjust = -0.1,
                +               size = 4) +
+     coord_flip() +
+     labs(
  +         title = "Average Units Purchased by Life Stage",
  +         x = "Life Stage",
  +         y = "Average Units"
  +     ) +
+     theme_minimal() +
+     theme(
  +         plot.title = element_text(face="bold",hjust=.5,size=16),
  +         axis.title = element_text(face="bold")
  +     )
```

8. Business Insight

Business Insights

1. Customer Segment Performance

The analysis revealed that Older Singles/Couples generated the highest total sales ($376,014), making them the most valuable customer segment. Retirees followed closely with total sales of $342,382 and also represented the largest customer base (14,555 customers). These findings indicate that older customer segments are the primary contributors to overall revenue.

2. Customer Type Performance

Among customer types, Mainstream customers generated the highest total sales ($700,865) and recorded the highest number of transactions (95,043). Budget customers ranked second, while Premium customers contributed the lowest overall sales. Although average spending per transaction was relatively similar across all customer types (approximately $7.30), the higher sales generated by Mainstream customers were driven by a larger customer base and more frequent purchases.

3. Purchasing Behaviour

The analysis showed that Older Families purchased the highest average number of units per transaction (1.95 units), followed closely by Young Families (1.94 units). This suggests that family households tend to buy larger quantities during each shopping trip, making them suitable targets for multi-buy promotions and family-sized product offerings.

4. Brand Performance

Kettle emerged as the highest-performing brand with total sales of approximately $390,240, significantly outperforming all other brands. It was also the top-selling brand across nearly every customer segment, indicating strong customer preference and consistent brand loyalty. Other high-performing brands included Smiths, Doritos, and Pringles.

5. Pack Size Preferences

The 175g pack size generated the highest total sales ($485,431) and the largest number of transactions (66,389), making it the most popular package size among customers. The 150g pack size ranked second, while larger packs such as 330g and 380g generated higher average spending per purchase but lower overall transaction volumes.

6. Sales Drivers

Overall sales performance was primarily driven by:

- High-value customer segments, particularly Older Singles/Couples and Retirees.
- Mainstream customers, who made the largest number of purchases.

- Strong consumer preference for the Kettle brand.
- High demand for the 175g pack size.
- Higher purchase quantities among family households.

These factors collectively contributed to higher transaction volumes and increased total sales.

Business Recommendations

Based on the findings, the following recommendations are proposed:

1. Increase inventory of Kettle products, as they consistently outperform competing brands across customer segments.
2. Prioritize stocking 175g pack sizes, which account for the highest sales and transaction volume.
3. Target Older Singles/Couples and Retirees through personalized marketing campaigns, loyalty programs, and promotional offers, as they represent the highest-value customer segments.
4. Develop bundle and multi-buy promotions aimed at Older Families and Young Families, who purchase the greatest number of units per transaction.
5. Focus marketing efforts on Mainstream customers, who contribute the highest overall revenue through frequent purchasing.

6. Leverage customer segmentation to tailor promotions based on life stage and purchasing behavior, improving marketing effectiveness and customer engagement.

9. Conclusion

This analysis successfully examined customer purchasing behavior by integrating transaction and customer data to identify key sales trends, customer preferences, and opportunities for business growth. Through data cleaning, feature engineering, exploratory data analysis, and visualization, valuable insights were generated to support evidence-based decision-making.

The findings revealed that Older Singles/Couples were the highest revenue-generating customer segment, while Retirees represented the largest customer base. Mainstream customers contributed the greatest overall sales, largely due to their higher purchase frequency. Product analysis showed that Kettle was the top-performing brand across customer segments, and the 175g pack size generated the highest sales and transaction volume. Additionally, family households purchased more units per transaction, indicating opportunities for quantity-based promotions.

Based on these insights, the client should focus on maintaining adequate inventory of high-performing products, particularly Kettle chips and 175g pack sizes, while implementing targeted marketing campaigns for high-value customer segments such as Older Singles/Couples, Retirees, and Mainstream customers. Promotional strategies such as bundle offers and loyalty programs may further strengthen customer engagement and increase sales.

Overall, the analysis demonstrates how customer segmentation and purchasing behavior can be leveraged to improve inventory management, optimize marketing strategies, and enhance overall business performance through data-driven decision-making.